
Linear filtering

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Roadmap

Machine Vision Technology							
Semantic information				Metric 3D information			
Pixels	Segments	Images	Videos	Camera		Multi-view Geometry	
Convolutions Edges & Fitting Local features Texture	Segmentation Clustering	Recognition Detection	Motion Tracking	Camera Model	Camera Calibration	Epipolar Geometry	SFM

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Types of Images

Binary



Gray Scale



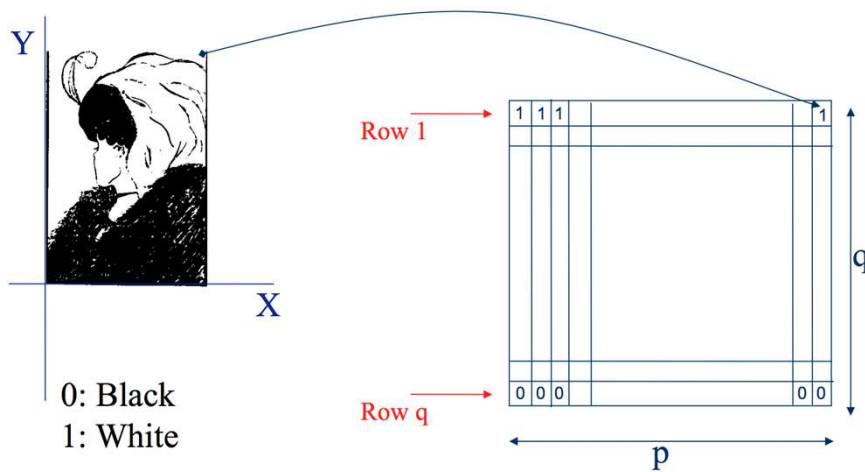
Color



Source: Ulas Bagci

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Binary image representation

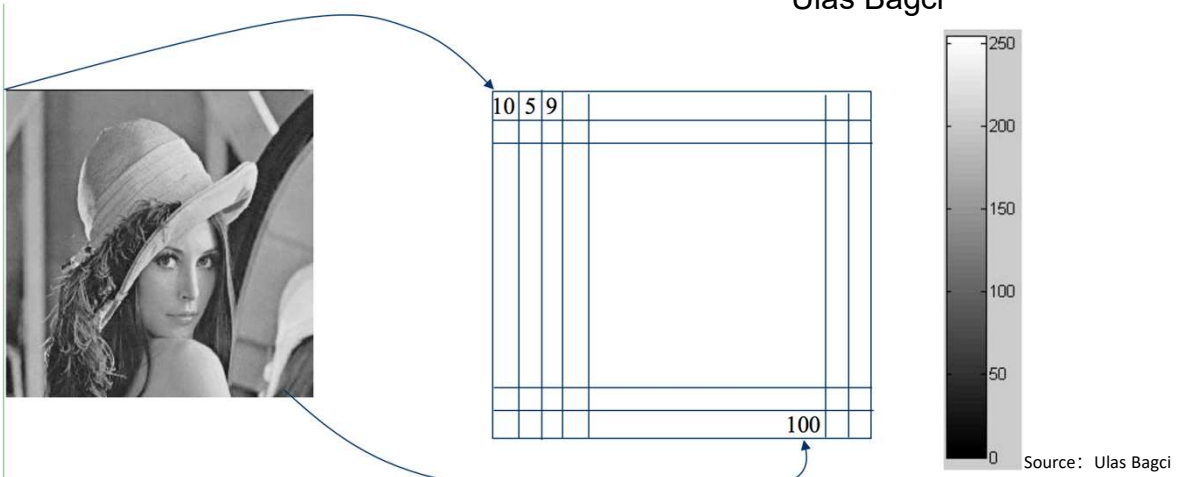


Source: Ulas Bagci

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Grayscale image representation

Slide credit:
Ulas Bagci



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Color Image - one channel



Source: Ulas Bagci

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Color image representation



Source: Ulas Bagci

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Motivation: Image denoising

- How can we reduce noise in a photograph?



Source: S. Lazebnik

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Moving average

- Let's replace each pixel with a *weighted* average of its neighborhood
- The weights are called the *filter kernel*
- What are the weights for the average of a 3x3 neighborhood?

$$\frac{1}{9}$$

1	1	1
1	1	1
1	1	1

"box filter"

Source: S. Lazebnik

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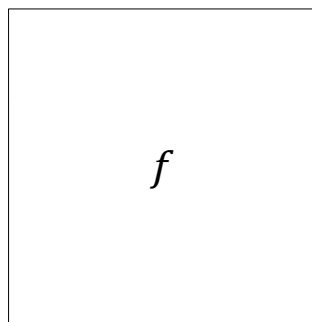
Defining convolution

- Let f be the image and g be the kernel. The output of convolving f with g is denoted $f * g$.

$$(f * g)[m, n] = \sum_{k, l} f[m - k, n - l] g[k, l]$$

b	u	!
p	e	j
a	q	c

Convention:
kernel is "flipped"



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Key properties

- **Linearity:** $\text{filter}(f_1 + f_2) = \text{filter}(f_1) + \text{filter}(f_2)$
- **Shift invariance:** same behavior regardless of pixel location: $\text{filter}(\text{shift}(f)) = \text{shift}(\text{filter}(f))$
- Theoretical result: any linear shift-invariant operator can be represented as a convolution

Source: S. Lazebnik

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Properties in more detail

- Commutative: $a * b = b * a$
 - Conceptually no difference between filter and signal
- Associative: $a * (b * c) = (a * b) * c$
 - Often apply several filters one after another: $((a * b_1) * b_2) * b_3$
 - This is equivalent to applying one filter: $a * (b_1 * b_2 * b_3)$
- Distributes over addition: $a * (b + c) = (a * b) + (a * c)$
- Scalars factor out: $ka * b = a * kb = k(a * b)$
- Identity: unit impulse $e = [\dots, 0, 0, 1, 0, 0, \dots]$,
 $a * e = a$

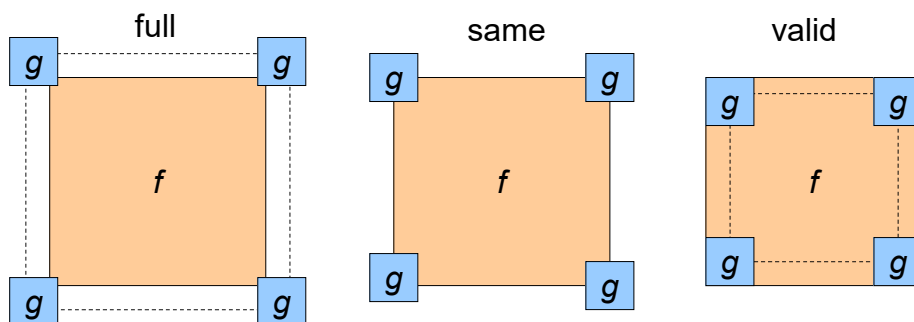
Source: S. Lazebnik

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Annoying details

What is the size of the output?

- MATLAB: `filter2(g, f, shape)`
 - `shape = 'full'`: output size is sum of sizes of f and g
 - `shape = 'same'`: output size is same as f
 - `shape = 'valid'`: output size is difference of sizes of f and g



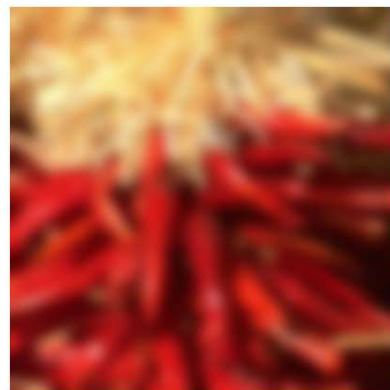
Source: S. Lazebnik

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Annoying details

What about near the edge?

- the filter window falls off the edge of the image
- need to extrapolate
- methods:
 - clip filter (black)
 - wrap around
 - copy edge
 - reflect across edge



Source: S. Marschner

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Annoying details

What about near the edge?

- the filter window falls off the edge of the image
- need to extrapolate
- methods (MATLAB):
 - clip filter (black): `imfilter(f, g, 0)`
 - wrap around: `imfilter(f, g, 'circular')`
 - copy edge: `imfilter(f, g, 'replicate')`
 - reflect across edge: `imfilter(f, g, 'symmetric')`

Source: S. Marschner

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Practice with linear filters



Original

0	0	0
0	1	0
0	0	0

?

Source: D. Lowe

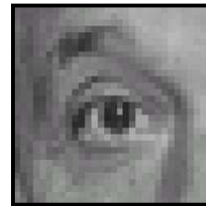
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Practice with linear filters



Original

0	0	0
0	1	0
0	0	0



Filtered
(no change)

Source: D. Lowe

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Practice with linear filters



Original

0	0	0
0	0	1
0	0	0

?

Source: D. Lowe

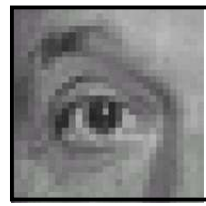
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Practice with linear filters



Original

0	0	0
0	0	1
0	0	0



Shifted *left*
By 1 pixel

Source: D. Lowe

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Practice with linear filters



Original

$\frac{1}{9}$	1	1	1
	1	1	1
	1	1	1

?

Source: D. Lowe

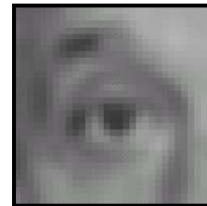
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Practice with linear filters



Original

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$



Blur (with a box filter)

Source: D. Lowe

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Practice with linear filters



Original

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

?

(Note that filter sums to 1)

Source: D. Lowe

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Practice with linear filters



Original

0	0	0
0	2	0
0	0	0

-

$\frac{1}{9}$

1	1	1
1	1	1
1	1	1



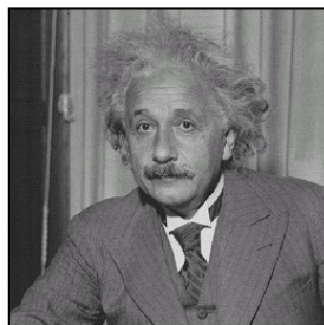
Sharpening filter

- Accentuates differences
with local average

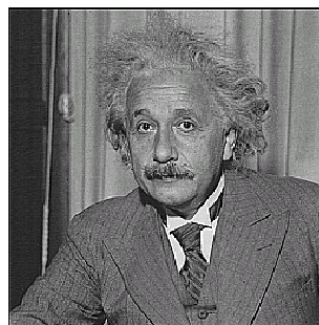
Source: D. Lowe

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Sharpening



before



after

Source: D. Lowe

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Sharpening

What does blurring take away?

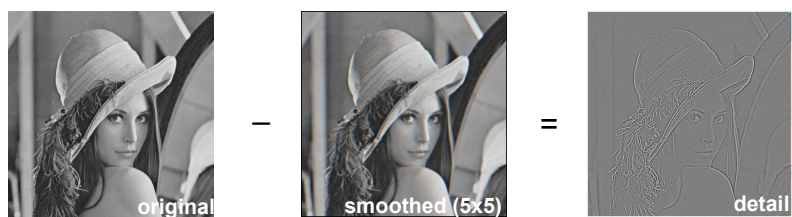


Source: D. Lowe

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Sharpening

What does blurring take away?



Let's add it back:

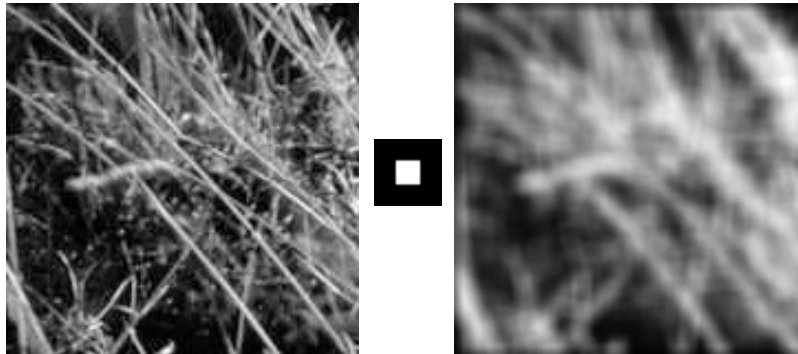


Source: D. Lowe

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Smoothing with box filter revisited

- What's wrong with this picture?
- What's the solution?

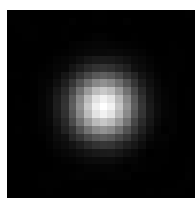


Source: D. Forsyth

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Smoothing with box filter revisited

- What's wrong with this picture?
- What's the solution?
 - To eliminate edge effects, weight contribution of neighborhood pixels according to their closeness to the center



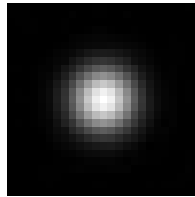
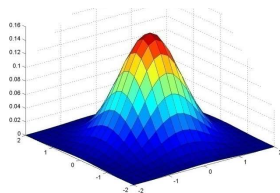
“fuzzy blob”

Source: S. Lazebnik

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Gaussian Kernel

$$G_{\sigma} = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$



0.003	0.013	0.022	0.013	0.003
0.013	0.059	0.097	0.059	0.013
0.022	0.097	0.159	0.097	0.022
0.013	0.059	0.097	0.059	0.013
0.003	0.013	0.022	0.013	0.003

5 x 5, $\sigma = 1$

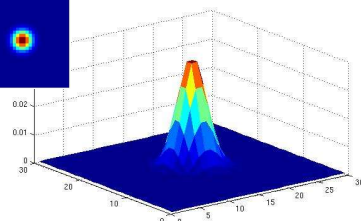
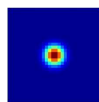
- Constant factor at front makes volume sum to 1 (can be ignored when computing the filter values, as we should renormalize weights to sum to 1 in any case)

Source: C. Rasmussen

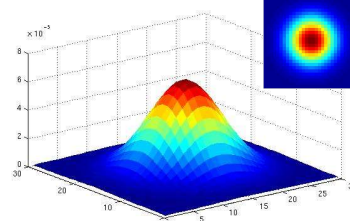
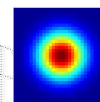
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Gaussian Kernel

$$G_{\sigma} = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$



$\sigma = 2$ with 30 x 30 kernel



$\sigma = 5$ with 30 x 30 kernel

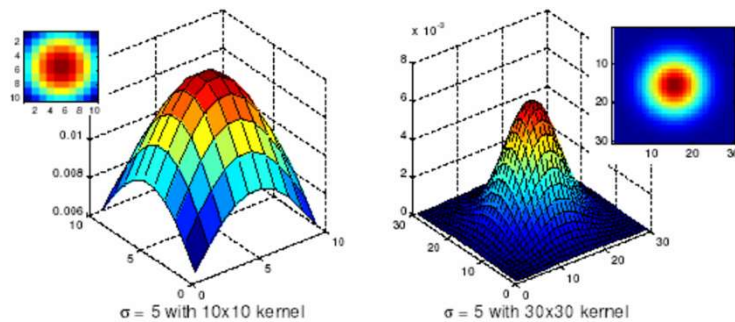
- Standard deviation σ : determines extent of smoothing

Source: K. Grauman

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Choosing kernel width

- The Gaussian function has infinite support, but discrete filters use finite kernels

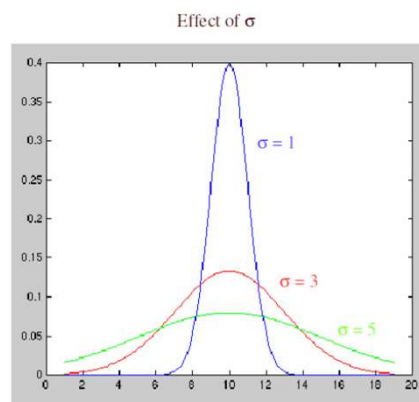


Source: K. Grauman

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Choosing kernel width

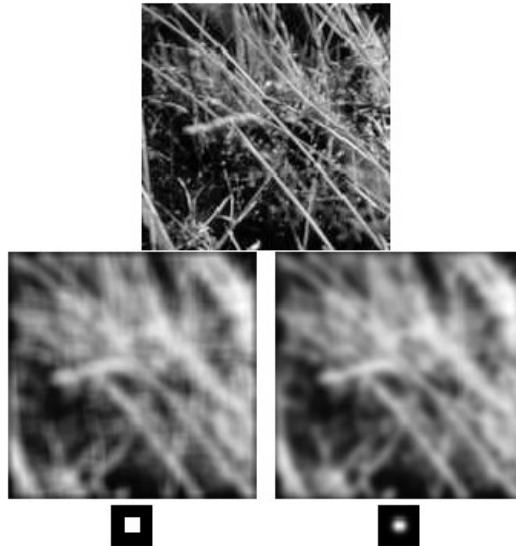
- Rule of thumb: set filter half-width to about 3σ



Source: S. Lazebnik

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Gaussian vs. box filtering



Source: S. Lazebnik

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Gaussian filters

- Remove “high-frequency” components from the image (low-pass filter)
- Convolution with self is another Gaussian
 - So can smooth with small- σ kernel, repeat, and get same result as larger- σ kernel would have
 - Convolving two times with Gaussian kernel with std. dev. σ is same as convolving once with kernel with std. dev. $\sigma\sqrt{2}$
- *Separable* kernel
 - Factors into product of two 1D Gaussians

Source: K. Grauman

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Separability of the Gaussian filter

$$\begin{aligned}
 G_{\sigma}(x, y) &= \frac{1}{2\pi\sigma^2} \exp^{-\frac{x^2 + y^2}{2\sigma^2}} \\
 &= \left(\frac{1}{\sqrt{2\pi}\sigma} \exp^{-\frac{x^2}{2\sigma^2}} \right) \left(\frac{1}{\sqrt{2\pi}\sigma} \exp^{-\frac{y^2}{2\sigma^2}} \right)
 \end{aligned}$$

The 2D Gaussian can be expressed as the product of two functions, one a function of x and the other a function of y

In this case, the two functions are the (identical) 1D Gaussian

Source: D. Lowe

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Separability example

2D convolution
(center location only)

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 3 & 3 \\ 3 & 5 & 5 \\ 4 & 4 & 6 \end{bmatrix}$$

The filter factors
into a product of 1D
filters:

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

Perform convolution
along rows:

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 3 & 3 \\ 3 & 5 & 5 \\ 4 & 4 & 6 \end{bmatrix} =$$

Followed by convolution
along the remaining column:

Source: K. Grauman

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Why is separability useful?

- What is the complexity of filtering an $n \times n$ image with an $m \times m$ kernel?
 - $O(n^2 m^2)$
- What if the kernel is separable?
 - $O(n^2 m)$

Source: S. Lazebnik

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Noise



Original



Salt and pepper noise



Impulse noise



Gaussian noise

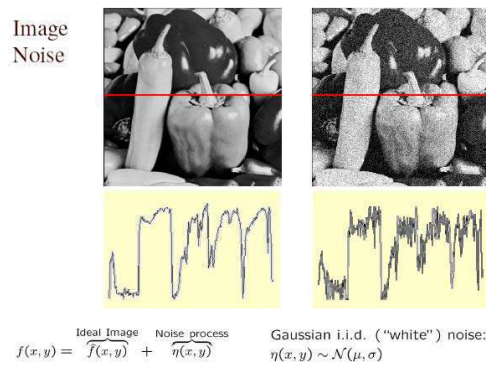
- **Salt and pepper noise:** contains random occurrences of black and white pixels
- **Impulse noise:** contains random occurrences of white pixels
- **Gaussian noise:** variations in intensity drawn from a Gaussian normal distribution

Source: S. Seitz

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Gaussian noise

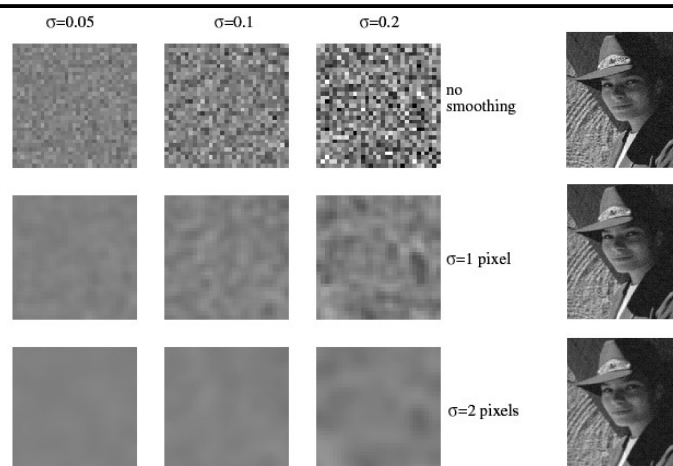
- Mathematical model: sum of many independent factors
- Good for small standard deviations
- Assumption: independent, zero-mean noise



Source: M. Hebert

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Reducing Gaussian noise



Smoothing with larger standard deviations suppresses noise, but also blurs the image

Source: S. Lazebnik

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Reducing salt-and-pepper noise



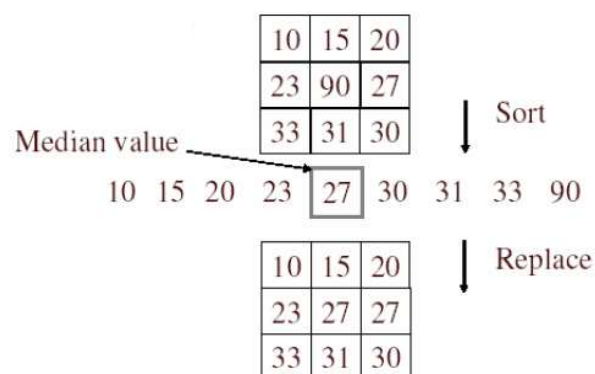
What's wrong with the results?

Source: S. Lazebnik

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Alternative idea: Median filtering

- A **median filter** operates over a window by selecting the median intensity in the window



- Is median filtering linear?

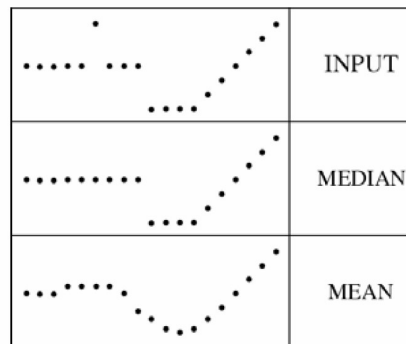
Source: K. Grauman

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Median filter

- What advantage does median filtering have over Gaussian filtering?
 - Robustness to outliers

filters have width 5 :

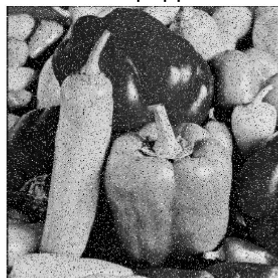


Source: K. Grauman

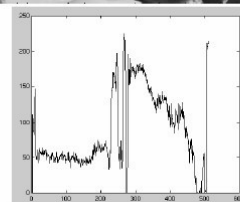
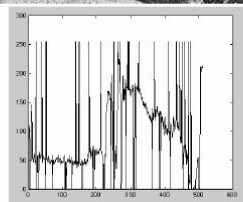
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Median filter

Salt-and-pepper noise



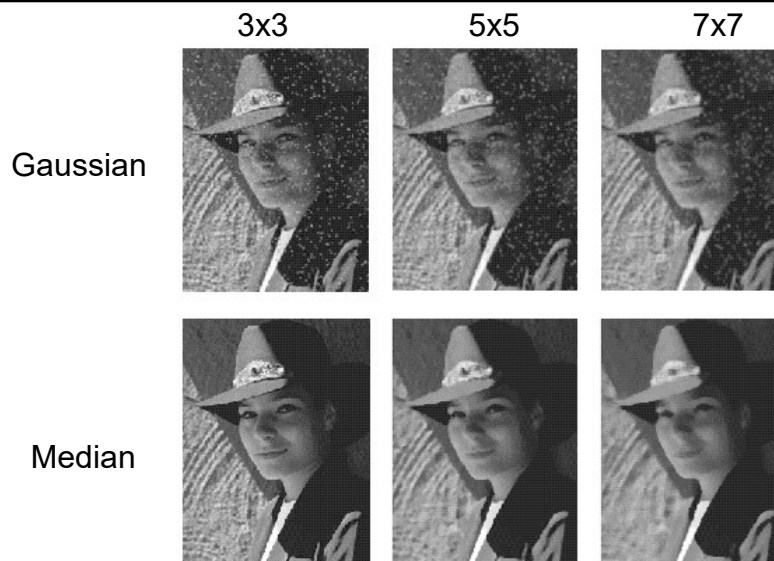
Median filtered



Source: M. Hebert

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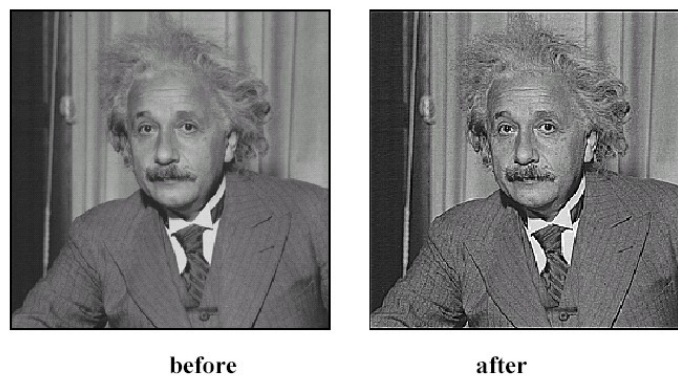
Gaussian vs. median filtering



Source: S. Lazebnik

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Sharpening revisited

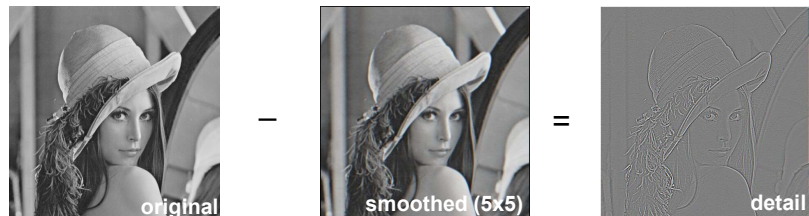


Source: D. Lowe

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Sharpening revisited

What does blurring take away?



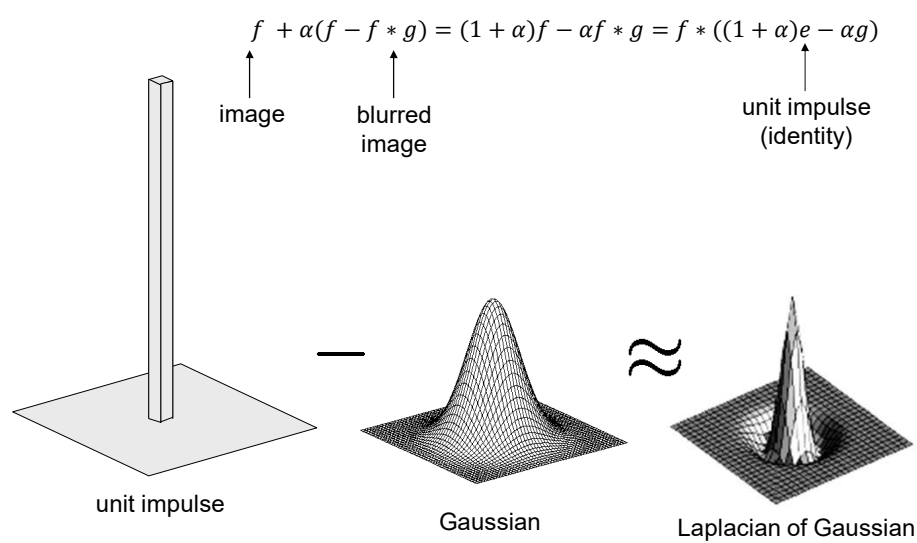
Let's add it back:



Source: S. Lazebnik

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Unsharp mask filter



Source: S. Lazebnik

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